

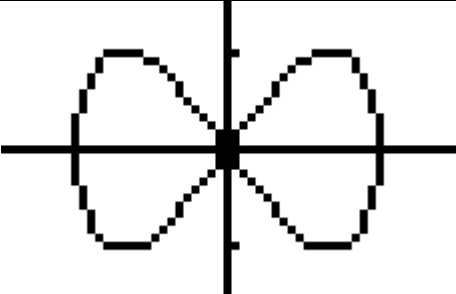
1i	$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ Since $\begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \neq m \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ for $m \in \mathbb{R}$, the lines are not parallel. $-2\lambda = 1 + \mu \quad \text{--- (1)}$ $\lambda = 1 - 2\mu \quad \text{---- (2)}$ $1 - \lambda = \mu \quad \text{----- (3)}$ Solving (1) and (2): $\lambda = -1, \mu = 1$ when $\lambda = 1, \mu = 0$, (1): LHS = $-2 \neq 1 = \text{RHS}$ Therefore the lines are skewed.
1ii	$\text{angle} = \cos^{-1} \frac{\left \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \right }{\sqrt{6}\sqrt{6}}$ $= 33.6^\circ$
2i	$\sin(A+B) + \sin(A-B)$ $= \sin A \cos B + \sin B \cos A + \sin A \cos B - \sin B \cos A$ $= 2 \sin A \cos B$
2ii	$AB + BC = \sin\left(\frac{\pi}{4} + \theta\right) + \sin\left(\frac{\pi}{4} - \theta\right)$ $= 2 \sin \frac{\pi}{4} \cos \theta$ $\approx 2 \left(\frac{\sqrt{2}}{2} \right) \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} \right)$ $= \sqrt{2} \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{24} \right)$ $p = \sqrt{2}, q = 24$

3i	$\frac{dx}{dt} = -kx$ $\int \frac{1}{x} dx = -k \int dt$ $\ln x = -kt + c$ when $t = 0$, $x = x_0$, $c = \ln x_0$ when $x = 0.5x_0$, $\ln 0.5x_0 = -kt + \ln x_0$ $t = -\frac{\ln 0.5}{k}$ $= \frac{\ln 2}{k}$
3ii	$\frac{\ln 2}{k} = 5730$ $k = \frac{\ln 2}{5730}$ when $x = 0.002x_0$, $\ln 0.002x_0 = -\frac{\ln 2}{5730}t + \ln x_0$ $t = -\frac{5730 \ln 0.002}{\ln 2}$ $= 51374 \text{ years}$
4	$\overrightarrow{OB} = \mathbf{a} + \mathbf{c} \text{ and } \overrightarrow{AC} = \mathbf{c} - \mathbf{a}$ $ \overrightarrow{OB} ^2 + \overrightarrow{AC} ^2 = \mathbf{a} + \mathbf{c} ^2 + \mathbf{c} - \mathbf{a} ^2$ $= (\mathbf{a} + \mathbf{c}) \cdot (\mathbf{a} + \mathbf{c}) + (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a})$ $= \mathbf{a} ^2 + 2\mathbf{a} \cdot \mathbf{c} + \mathbf{c} ^2 + \mathbf{a} ^2 - 2\mathbf{a} \cdot \mathbf{c} + \mathbf{c} ^2$ $= \overrightarrow{OA} ^2 + \overrightarrow{AB} ^2 + \overrightarrow{BC} ^2 + \overrightarrow{CO} ^2$ Alternative solution: $ \overrightarrow{OA} ^2 + \overrightarrow{AB} ^2 + \overrightarrow{BC} ^2 + \overrightarrow{CO} ^2 = \mathbf{a} ^2 + \mathbf{c} ^2 + \mathbf{a} ^2 + \mathbf{c} ^2$ $= \mathbf{a} ^2 + 2\mathbf{a} \cdot \mathbf{c} + \mathbf{c} ^2 + \mathbf{a} ^2 - 2\mathbf{a} \cdot \mathbf{c} + \mathbf{c} ^2$ $= (\mathbf{a} + \mathbf{c}) \cdot (\mathbf{a} + \mathbf{c}) + (\mathbf{c} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{a})$ $= \mathbf{a} + \mathbf{c} ^2 + \mathbf{c} - \mathbf{a} ^2$ $= \overrightarrow{OB} ^2 + \overrightarrow{AC} ^2$
4	$ \overrightarrow{OA} ^2 + \overrightarrow{AB} ^2 + \overrightarrow{BC} ^2 + \overrightarrow{CO} ^2 = \overrightarrow{OB} ^2 + \overrightarrow{AC} ^2$ $ \overrightarrow{OA} ^2 + \overrightarrow{OC} ^2 = \overrightarrow{OB} ^2$ $ \mathbf{a} ^2 + \mathbf{c} ^2 = \mathbf{a} + \mathbf{c} ^2$ $= \mathbf{a} ^2 + 2\mathbf{a} \cdot \mathbf{c} + \mathbf{c} ^2$ $\mathbf{a} \cdot \mathbf{c} = 0$ $\mathbf{a}$ is perpendicular to $\mathbf{c}$.

5ai	Since T_n is an AP, $T_n = a + (n-1)d.$ $\frac{e^{T_{n+1}}}{e^{T_n}} = e^d \text{ which is still a constant,}$ $\therefore$ the terms of the sequence with nth term e^{T_n} follows a GP.
5aii	Since X_n is a GP, $X_n = ar^{n-1}.$ $\ln X_{n+1} - \ln X_n = \ln ar^n - \ln ar^{n-1}$ $= \ln r, \text{ which is a constant}$ $\therefore$ the terms of the sequence with nth term $\ln X_n$ follows an AP.
5b	$\frac{a+2d}{a} = \frac{a+3d}{a+2d}$ $a^2 + 4ad + 4d^2 = a^2 + 3ad$ $d(a+4d) = 0$ $d = 0 \text{ (NA) or } a = -4d$ $r = \frac{-4d+2d}{-4d}$ $= \frac{1}{2}$ Since $r < 1$, GP is convergent.

6i	$\frac{d^2y}{dx^2} = ae^{2x} + be^{-3x}$ $\frac{dy}{dx} = \frac{a}{2}e^{2x} - \frac{b}{3}e^{-3x} + c$ $y = \frac{a}{4}e^{2x} + \frac{b}{9}e^{-3x} + cx + d$
6ii	From Maclaurin series, when $x=0$, $y=2$, $\frac{dy}{dx}=-1$, $\frac{d^2y}{dx^2}=13$: $\frac{a}{4} + \frac{b}{9} + 0c + d = 2$ $\frac{a}{2} - \frac{b}{3} + c = -1$ $a + b + 0c + 0d = 13$ Alternative solution: $y = \frac{a}{4}e^{2x} + \frac{b}{9}e^{-3x} + cx + d$ $\approx \frac{a}{4}\left(1 + 2x + \frac{4x^2}{2}\right) + \frac{b}{9}\left(1 - 3x + \frac{9x^2}{2}\right) + cx + d$ $2 - x + \frac{13}{2}x^2 \approx \left(\frac{a}{4} + \frac{b}{9} + d\right) + \left(\frac{a}{2} - \frac{b}{3} + c\right)x + \left(\frac{a}{2} + \frac{b}{2}\right)x^2$ $\frac{a}{4} + \frac{b}{9} + d = 2$ $\frac{a}{2} - \frac{b}{3} + c = -1$ $\frac{a}{2} + \frac{b}{2} = \frac{13}{2}$ Curve passes through $\left(\ln 2, \frac{33}{8}\right)$: $a + \frac{b}{72} + c \ln 2 + d = \frac{33}{8}$ Using GC, $a=4$, $b=9$, $c=d=0$. $\therefore y = e^{2x} + e^{-3x}$

7a	$\int \tan^{-1} x \, dx = x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{x^2 + 1} \, dx + c'$ $= x \tan^{-1} x - \frac{1}{2} \ln(x^2 + 1) + c$
7b	$\int \frac{2x}{x^2 + 2x + 1} \, dx = \int \frac{2x + 2}{x^2 + 2x + 1} - \frac{2}{(x + 1)^2} \, dx$ $= \ln(x^2 + 2x + 1) + \frac{2}{x + 1} + c$ Alternative solution: $\int \frac{2x}{x^2 + 2x + 1} \, dx = \int \frac{2}{x + 1} - \frac{2}{(x + 1)^2} \, dx$ $= 2 \ln(x + 1) + \frac{2}{x + 1} + c$
7c	$\int_1^2 \frac{1}{x\sqrt{x^2 - 1}} \, dx = \int_1^{\frac{1}{2}} \frac{1}{\frac{1}{u}\sqrt{\frac{1}{u^2} - 1}} \left(-\frac{1}{u^2}\right) \, du$ $= \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{1 - u^2}} \, du$ $= \left[\sin^{-1} u \right]_{\frac{1}{2}}^1$ $= \frac{\pi}{3}$ Alternative solution: $\int_1^2 \frac{1}{x\sqrt{x^2 - 1}} \, dx = \int_1^{\frac{1}{2}} \frac{1}{\frac{1}{u}\sqrt{\frac{1}{u^2} - 1}} \left(-\frac{1}{u^2}\right) \, du$ $= \int_1^{\frac{1}{2}} -\frac{1}{\sqrt{1 - u^2}} \, du$ $= \left[\cos^{-1} u \right]_{\frac{1}{2}}^1$ $= \frac{\pi}{3}$

8i	
8ii	Since the figure is symmetrical, $Area = 4 \int_{\frac{\pi}{2}}^0 (\sin 2t)(-\sin t) dt$ $= 4 \int_0^{\frac{\pi}{2}} \sin t \sin 2t dt$ $= 8 \int_0^{\frac{\pi}{2}} \sin^2 t \cos t dt$ $Area = 8 \left[\frac{\sin^3 t}{3} \right]_0^{\frac{\pi}{2}}$ $= \frac{8}{3}$
8iii	$y = \sin 2t$ $= 2 \sin t \cos t$ $= \pm 2x\sqrt{1-x^2}$ $Volume = 2\pi \int_0^1 \left(2x\sqrt{1-x^2} \right)^2 dx$ $= 3.351$ Alternatively, $Volume = 2\pi \int_{\frac{\pi}{2}}^0 (\sin 2t)^2 (-\sin t) dt$ $= 3.351$

9i	The figure consists of two vertically stacked coordinate planes. The top graph shows the intersection of the parabola $y = x^2$ and the rational function $y = \frac{x-1}{x-2}$. The parabola is shown in grey, and the rational function is shown in blue. They intersect at two points: $(-0.802, 0.643)$ and $(2.25, 5.05)$. The rational function has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 1$. The bottom graph shows the intersection of the parabola $y = x^2$ and the rational function $y^2 = \frac{x-1}{x-2}$. The parabola is shown in grey, and the rational function is shown in purple. They intersect at three points: $(-0.802, -0.802)$, $(0.555, 0.555)$, and $(2.25, 2.25)$. The rational function has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 1$.	
9ii	$x^3 - 2x^2 - x + 1 = 0$ $x^2(x - 2) = x - 1$ $x^2 = \frac{x-1}{x-2} = y \text{ for } C_1$ Alternatively, $x^2 = \frac{x-1}{x-2} = y \text{ for } C_1$ $x^2(x - 2) = x - 1$ $x^3 - 2x^2 - x + 1 = 0$ The solutions to the simultaneous equations $y = x^2$ and $y = \frac{x-1}{x-2}$ give the x coordinates of points of intersection between C_1 and $y = x^2$. Refer to graph of C_1. $x = -0.802, 0.555, 2.25$	
9iii	$x^2 = \frac{x-1}{x-2} = y \text{ for } C_1$ $x = \pm \sqrt{\frac{x-1}{x-2}} = y \text{ for } C_2$ $\therefore x = -0.802, 0.555, 2.25$	

10i	$(1-i)^n = \left[\sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right) \right]^n$ $= 2^{\frac{n}{2}} \left(\cos\left(-\frac{n\pi}{4}\right) + i \sin\left(-\frac{n\pi}{4}\right) \right)$ $= 2^{\frac{n}{2}} \left(\cos\frac{n\pi}{4} - i \sin\frac{n\pi}{4} \right)$ Alternatively, $(1-i)^n = \left[\sqrt{2} e^{-i\frac{\pi}{4}} \right]^n$ $= \sqrt{2}^n e^{-i\frac{n\pi}{4}}$ $= 2^{\frac{n}{2}} \left(\cos\frac{n\pi}{4} - i \sin\frac{n\pi}{4} \right)$ Alternatively, $ 1-i ^n = \sqrt{2}^n$ $= 2^{\frac{n}{2}}$ $\arg(1-i)^n = n \arg(1-i)$ $= -n \frac{\pi}{4}$ $\therefore (1-i)^n = 2^{\frac{n}{2}} \left(\cos\frac{n\pi}{4} - i \sin\frac{n\pi}{4} \right)$
10ii	$(1-i)^n = 2^{\frac{n}{2}} \left(\cos\frac{n\pi}{4} - i \sin\frac{n\pi}{4} \right)$ $\sin\frac{n\pi}{4} = 0 \qquad \cos\frac{n\pi}{4} < 0$ $\frac{n\pi}{4} = k\pi \qquad \frac{\pi}{2} < \frac{n\pi}{4} < \frac{3\pi}{2}$ $n = 4k, k \in \mathbb{Z}^+ \qquad 2 < n < 6$ $\therefore n = 4$ Otherwise... $(1-i)^n = 2^{\frac{n}{2}} \left(\cos\frac{n\pi}{4} - i \sin\frac{n\pi}{4} \right) = 2^{\frac{n}{2}} e^{-i\frac{n\pi}{4}}$ $-\frac{n\pi}{4} = (2k+1)\pi$ $n = -4(2k+1), k \in \mathbb{Z}^-$ $\therefore n = 4$
10iii	Since the <u>coefficients of the equation are real</u>, <u>complex roots</u> will occur in <u>conjugate pairs</u> by Conjugate Roots Theorem. Furthermore, the order of the equation is 4, we would expect 4 roots ie. <u>0 pair of complex root with 4 real roots</u> or <u>1 pair of complex conjugate roots with 2 real roots</u> or <u>2 pairs of complex conjugate roots and no real root</u>.
10iv	$z^4 + 4 = 0$ $z^4 = -4 = 4e^{i(-\pi+2k\pi)}, k = 0, \pm 1, -2$ $z = \sqrt[4]{4} e^{i\left(\frac{\pi}{4} + \frac{k\pi}{2}\right)}, k = 0, \pm 1, -2$

$$\text{when } k=0, z = \sqrt{2}e^{i\left(\frac{\pi}{4}\right)} = 1+i$$

$$\text{when } k=-1, z = \sqrt{2}e^{i\left(-\frac{\pi}{4}\right)} = 1-i$$

$$\text{when } k=1, z = \sqrt{2}e^{i\left(\frac{3\pi}{4}\right)} = -1+i$$

$$\text{when } k=-2, z = \sqrt{2}e^{i\left(-\frac{3\pi}{4}\right)} = -1-i$$

Alternatively:

$$z^4 + 4 = 0$$

$$(z^2 - 2i)(z^2 + 2i) = 0$$

$$z = \frac{\pm\sqrt{-4(-2i)}}{2} = \pm(1+i)$$

$$z = \frac{\pm\sqrt{-4(2i)}}{2} = \pm(1-i)$$

Alternatively:

Let $z = x + iy$.

$$(x + iy)^4 + 4 = 0$$

$$x^4 + 4ix^3y - 6x^2y^2 - 4ixy^3 + y^4 + 4 = 0$$

$$x^4 - 6x^2y^2 + y^4 + 4 = 0 \quad \text{and} \quad 4x^3y - 4xy^3 = 0$$

$$4xy(x^2 - y^2) = 0$$

$$x = \pm y$$

when $x = y$, when $x = -y$,

$$y^4 = 1 \qquad y^4 = 1$$

$$y = \pm 1 \qquad y = \pm 1$$

$$x = \pm 1 \qquad x = \mp 1$$

$$\therefore z = 1+i, -1-i, 1-i, -1+i$$

11i	$\frac{1}{x_{r-1}x_r} - \frac{1}{x_r x_{r+1}} = \frac{x_{r+1} - x_{r-1}}{x_{r-1}x_r x_{r+1}}$ $= \frac{2x_r}{x_{r-1}x_r x_{r+1}}$ $= \frac{2}{x_{r-1}x_{r+1}}$
11ii	$\sum_{r=2}^n \frac{1}{x_{r-1}x_{r+1}} = \frac{1}{2} \sum_{r=2}^n \left(\frac{1}{x_{r-1}x_r} - \frac{1}{x_r x_{r+1}} \right)$ $= \frac{1}{2} \left(\frac{1}{x_1 x_2} - \frac{1}{x_2 x_3} \right.$ $+ \frac{1}{x_2 x_3} - \frac{1}{x_3 x_4}$ $+ \dots$ $\left. + \frac{1}{x_{n-1}x_n} - \frac{1}{x_n x_{n+1}} \right)$ $= \frac{1}{2} \left(1 - \frac{1}{x_n x_{n+1}} \right)$
11iii	Let P_n be the statement $\sum_{r=2}^n \frac{1}{x_{r-1}x_{r+1}} = \frac{1}{2} \left(1 - \frac{1}{x_n x_{n+1}} \right)$ for $n \in \mathbb{N}^+$, $n \geq 2$. when $n = 2$, $\text{LHS} = \sum_{r=2}^2 \frac{1}{x_{r-1}x_{r+1}} = \frac{1}{x_1 x_3} = \frac{1}{1 \cdot 3} = \frac{1}{3}$ $\text{RHS} = \frac{1}{2} \left(1 - \frac{1}{x_2 x_3} \right) = \frac{1}{2} \left(1 - \frac{1}{1 \cdot 3} \right) = \frac{1}{3} = \text{LHS}$ $\therefore P_2$ is true. Assume P_k true for some $k \in \mathbb{N}^+$, $k \geq 2$ ie $\sum_{r=2}^k \frac{1}{x_{r-1}x_{r+1}} = \frac{1}{2} \left(1 - \frac{1}{x_k x_{k+1}} \right)$ To prove P_{k+1} is true $\sum_{r=2}^{k+1} \frac{1}{x_{r-1}x_{r+1}} = \frac{1}{2} \left(1 - \frac{1}{x_{k+1}x_{k+2}} \right)$ when $n = k + 1$, $\text{LHS} = \frac{1}{2} \left(1 - \frac{1}{x_k x_{k+1}} \right) + \frac{1}{x_k x_{k+2}}$ $= \frac{1}{2} \left(1 - \frac{1}{x_k x_{k+1}} + \frac{1}{x_k x_{k+1}} - \frac{1}{x_{k+1}x_{k+2}} \right)$ $= \frac{1}{2} \left(1 - \frac{1}{x_{k+1}x_{k+2}} \right) = \text{RHS}$ P_k true $\Rightarrow P_{k+1}$ true. Since P_2 true and P_k true $\Rightarrow P_{k+1}$ true, P_n true for all $n \in \mathbb{N}^+$, $n \geq 2$.
11iv	Since $x_n \rightarrow \infty$ as $n \rightarrow \infty$, $\frac{1}{x_n x_{n+1}} \rightarrow 0 \text{ and } \sum_{r=2}^n \frac{1}{x_{r-1}x_{r+1}} \rightarrow \frac{1}{2} \text{ as } n \rightarrow \infty.$

12i	$\overrightarrow{AB} = \begin{pmatrix} -9+3\lambda \\ 15-7\lambda \\ -4+4\lambda \end{pmatrix}$ $ \overrightarrow{AB} ^2 = (-9+3\lambda)^2 + (15-7\lambda)^2 + (-4+4\lambda)^2 = 10^2$ $\lambda^2 - 4\lambda + 3 = 0$ $\lambda = 3 \text{ or } \lambda = 1$ $\overrightarrow{OB} = \begin{pmatrix} 2 \\ -6 \\ 7 \end{pmatrix} \quad \overrightarrow{OC} = \begin{pmatrix} -4 \\ 8 \\ -1 \end{pmatrix}$
12ii	$\overrightarrow{OM} = \frac{1}{2} \left[\begin{pmatrix} 2 \\ -6 \\ 7 \end{pmatrix} + \begin{pmatrix} -4 \\ 8 \\ -1 \end{pmatrix} \right] = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ $\mathbf{n} = \overrightarrow{AM} = \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix}$ $\mathbf{r} \cdot \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -7 \\ 15 \\ -5 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \\ 4 \end{pmatrix}$ $-3x + y + 4z = 21 + 15 - 20$ $= 16$
12iii	Since $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} = 5$ and $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 1$, A lies in both π_1 and π_2.
12iv	$-3x + y + 4z = 16$ $x + 2y - 3z = 5$ $x - 2y + z = 1$ Using GC, $x = 15$, $y = 13$, $z = 12$ $\overrightarrow{OD} = \begin{pmatrix} 15 \\ 13 \\ 12 \end{pmatrix}$
12v	perpendicular height $= \overrightarrow{AM}$ $= \sqrt{9+1+16}$ $= \sqrt{26}$

$$\overrightarrow{BD} = \begin{pmatrix} 13 \\ 19 \\ 5 \end{pmatrix}, \quad \overrightarrow{CD} = \begin{pmatrix} 19 \\ 5 \\ 13 \end{pmatrix}$$

$$\text{area of base} = \frac{1}{2} |\overrightarrow{BD} \times \overrightarrow{CD}|$$

$$= \frac{1}{2} \left| \begin{pmatrix} 222 \\ -74 \\ -296 \end{pmatrix} \right|$$

$$= \sqrt{35594}$$

$$\text{volume} = \frac{1}{3} \times \sqrt{26} \times \sqrt{35594}$$

$$= \frac{962}{3} \text{ units}^3$$